



ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

Preliminaries

Stefano Ferretti

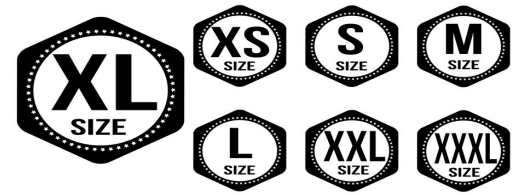
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Some terms used in these slides

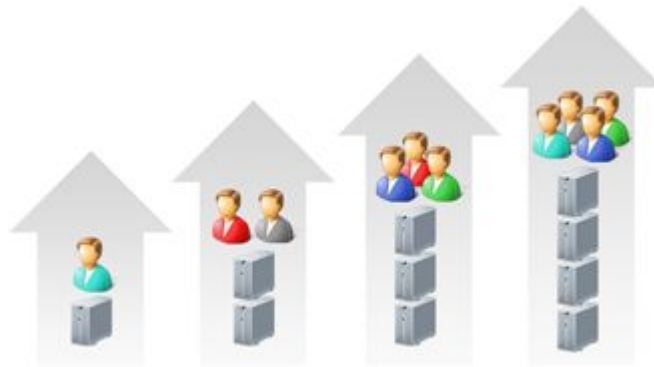
- Scalability
- Availability
- Single point of failure
- Bottleneck
- Churn
- Distributed architectures
 - client/server
 - peer-to-peer



Scalability



- Wikipedia: Scalability is the capability of a system, network, or process to handle a growing amount of work, or its potential to be enlarged in order to accommodate that growth



Availability



- Availability means the probability that a system is **operational at a given time**
 - the amount of time a system is actually operating as the percentage of total time it should be operating
- Availability features allow the system to stay operational even when faults do occur

AVAILABILITY: $\frac{\text{OPERATING TIME}}{\text{CALENDAR TIME}}$

UPTIME: $\frac{\text{OPERATING TIME}}{\text{SCHEDULED TIME}}$



Availability



$A = \dots$

$N = \dots$

MTTR

MTBF

99.999 %

Availability

Non-Availability

Mean Time To Repair

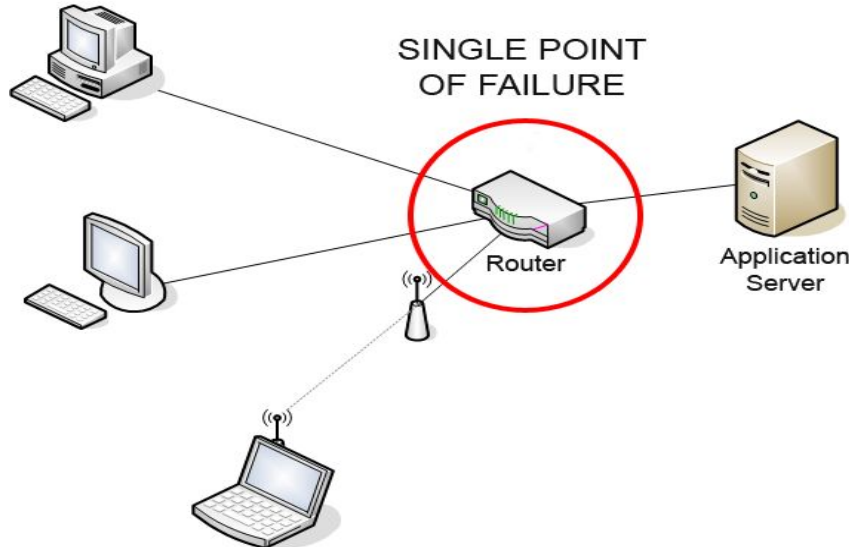
Mean Time Between Failures

$$A = \text{MTBF} / (\text{MTBF} + \text{MTTR})$$

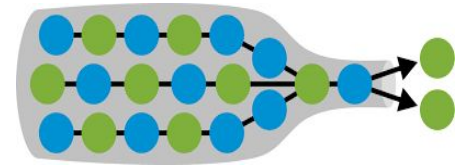


Single point of failure

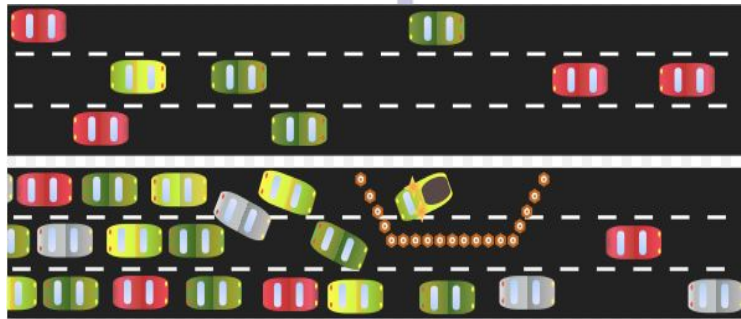
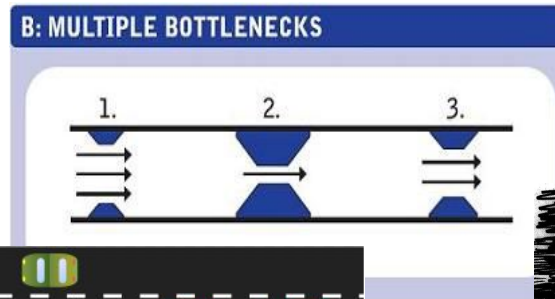
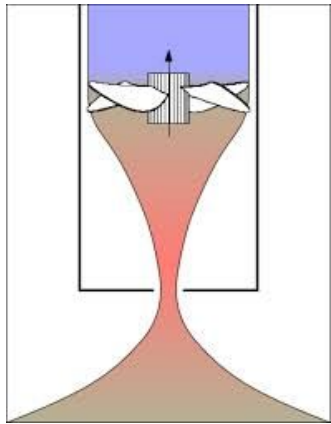
- A single point of failure is a part of a system that, if it fails, will stop the entire system from working



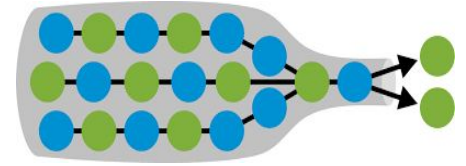
Bottleneck



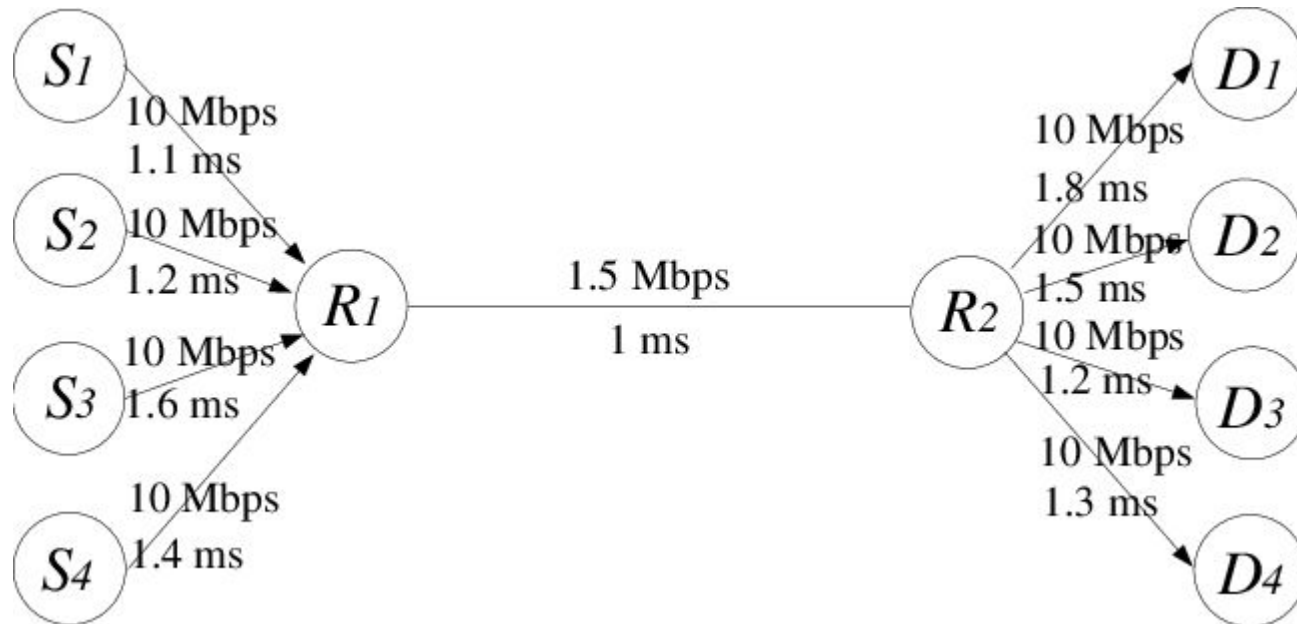
- A bottleneck is a point of congestion in a system that occurs when workloads arrive at a given point more quickly than that point can handle them
- The inefficiencies brought about by the bottleneck often create a queue and a longer overall cycle time



Bottleneck



- An example with communication networks

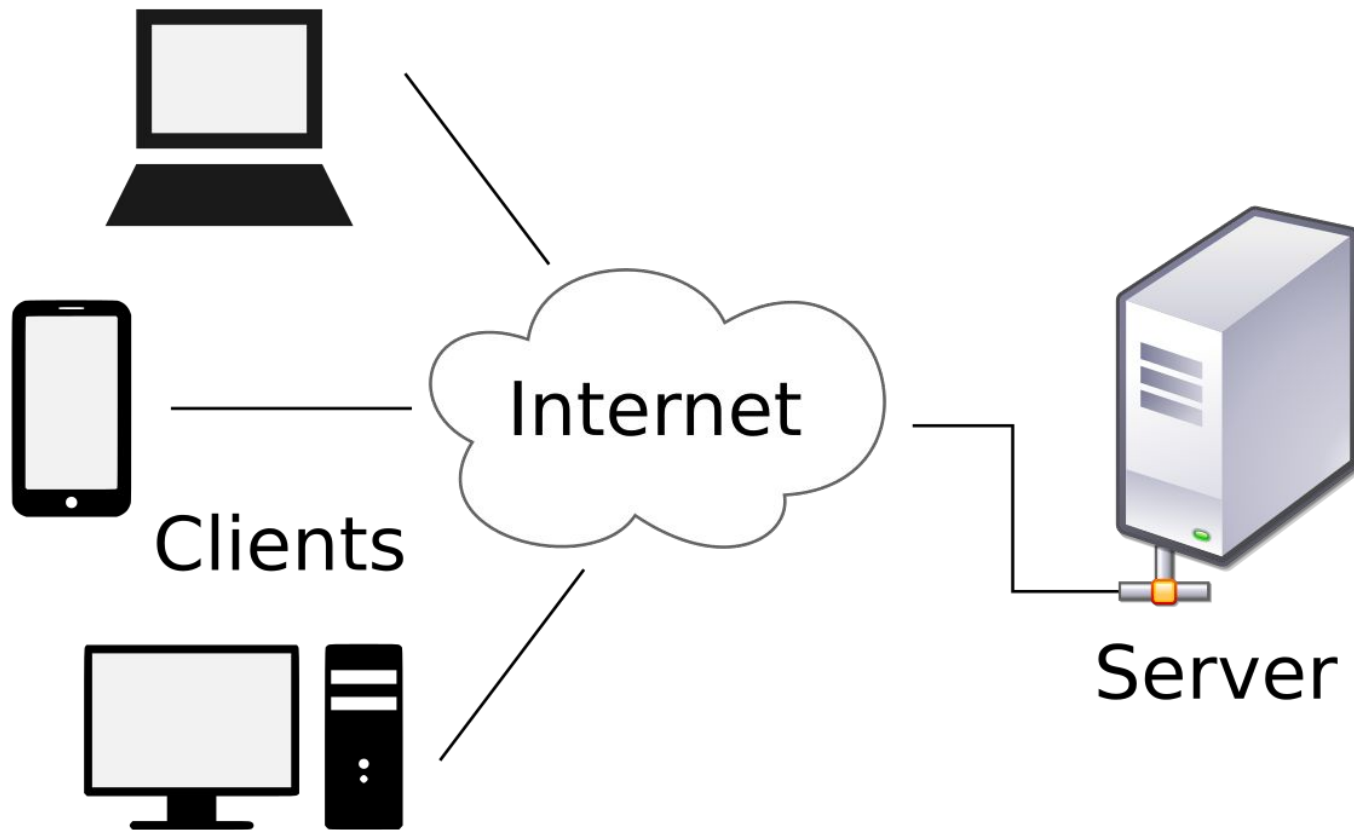


Two Main Typologies of Distributed Systems

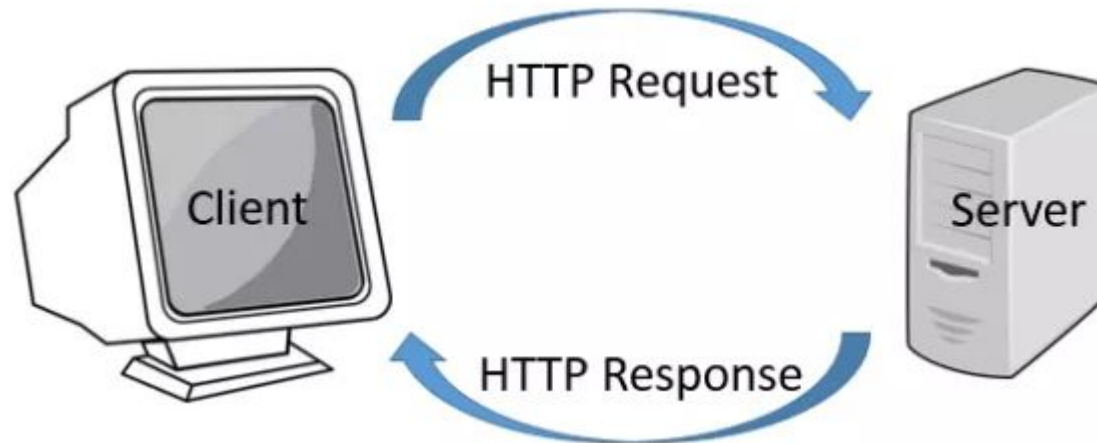
- Client/server
- Peer-to-peer



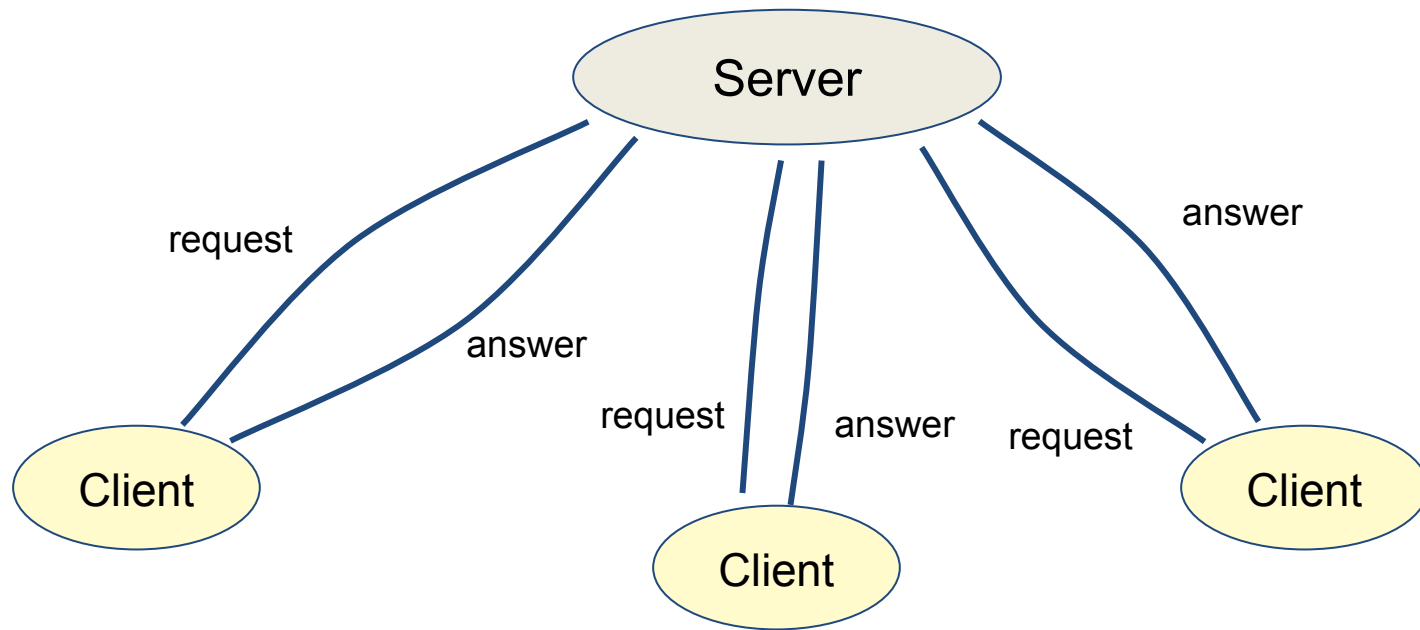
Client/Server Model



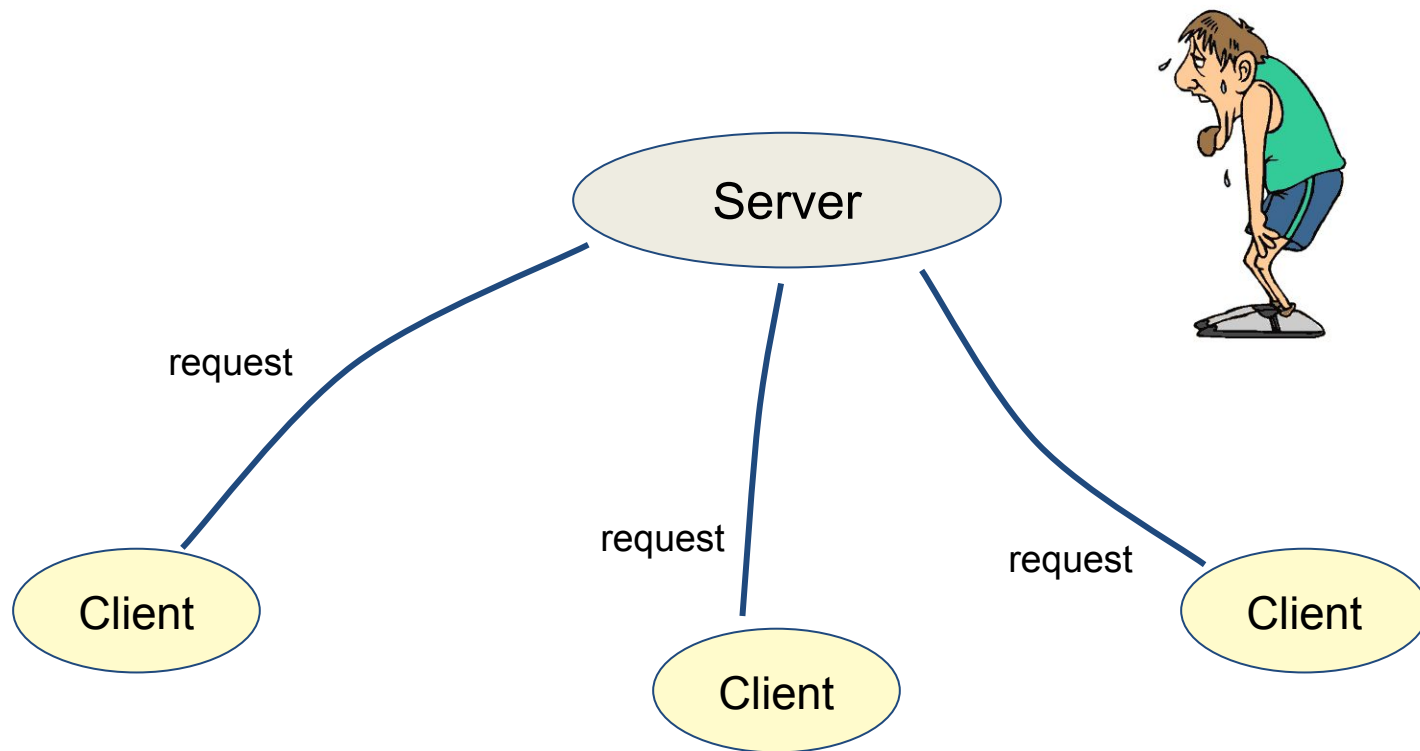
Example of C/S: the Web



Client/Server Model



Client/Server Model

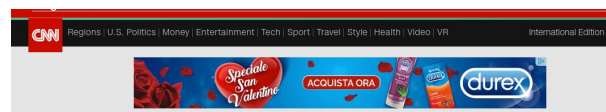
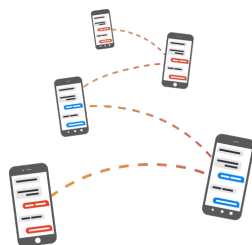
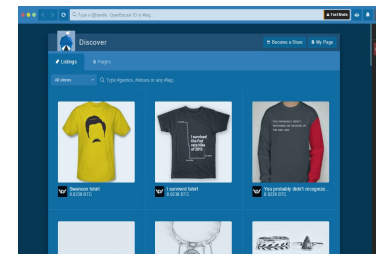
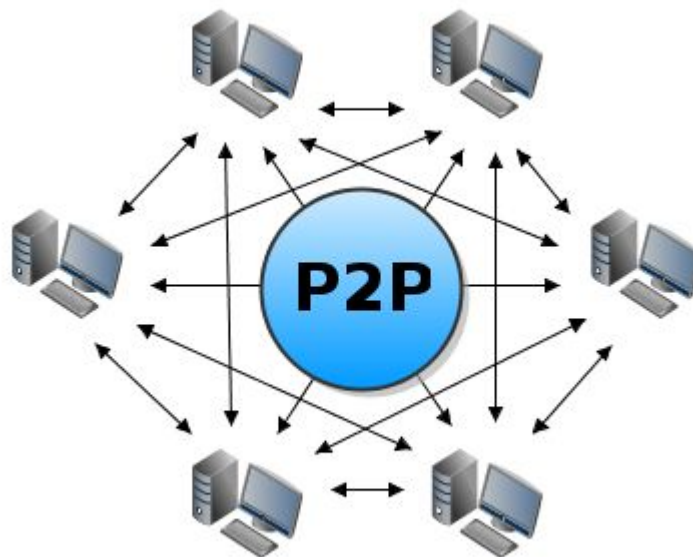
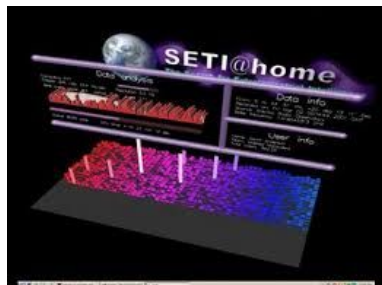


Churn

- Independent arrival and departure by a “large” number of peers
- Often hard to handle

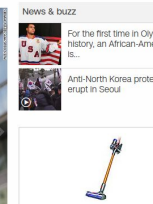


Peer-to-Peer Model



FireChat in Hong Kong: How an app tapped its way into the protests

By Peter Shadbolt, for CNN
Updated 1001 GMT (1801 HKT) October 16, 2014

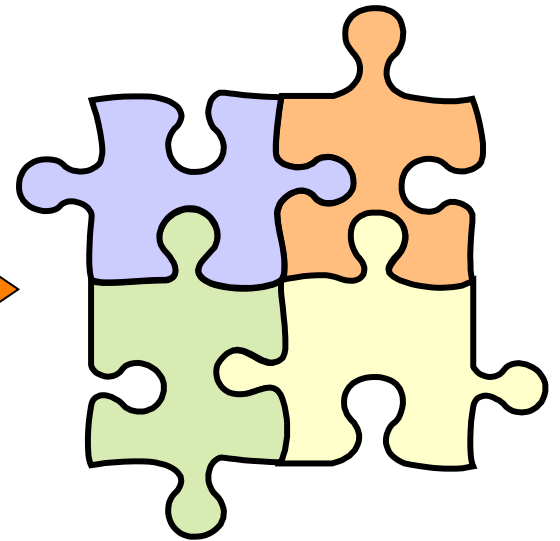
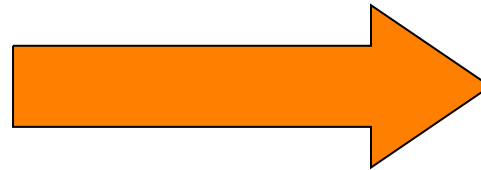
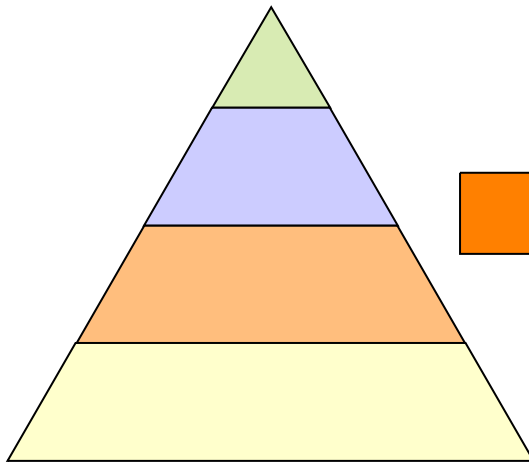
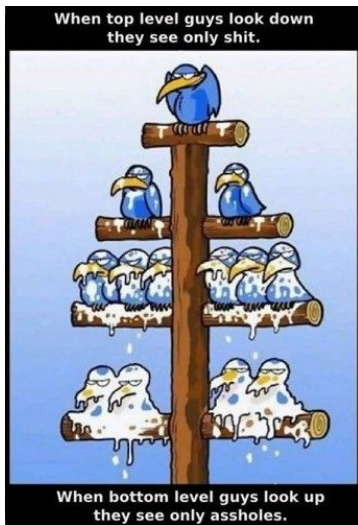


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How P2P changes the picture

P2P networks replace centralization and hierarchy with distribution and collaboration

At a philosophical level they replace centralized control with responsibility and freedom



The P2P concept in images



The P2P concept in images



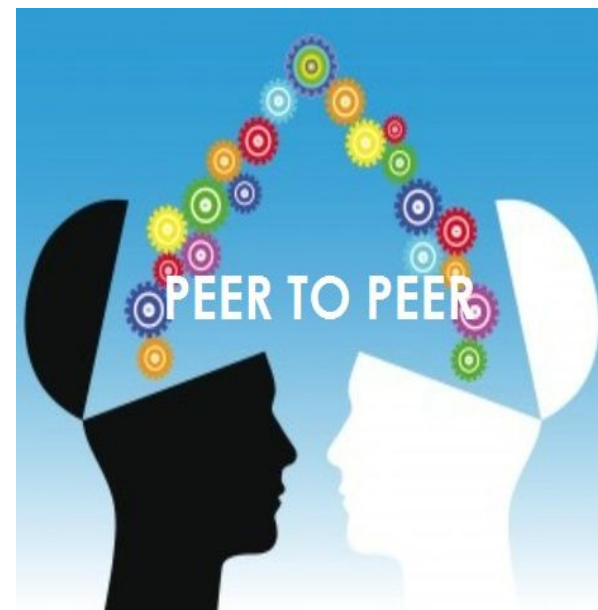
The P2P concept in images



The P2P concept in images

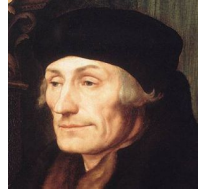


The P2P concept in images

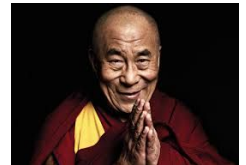


Quotes about sharing

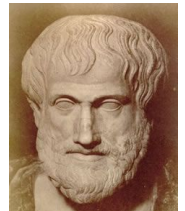
"There is no joy in possession without sharing." –
Erasmus



"Share your knowledge. It is a way to achieve
immortality." – Dalai Lama



"There is no delight in owning anything unshared." –
Seneca



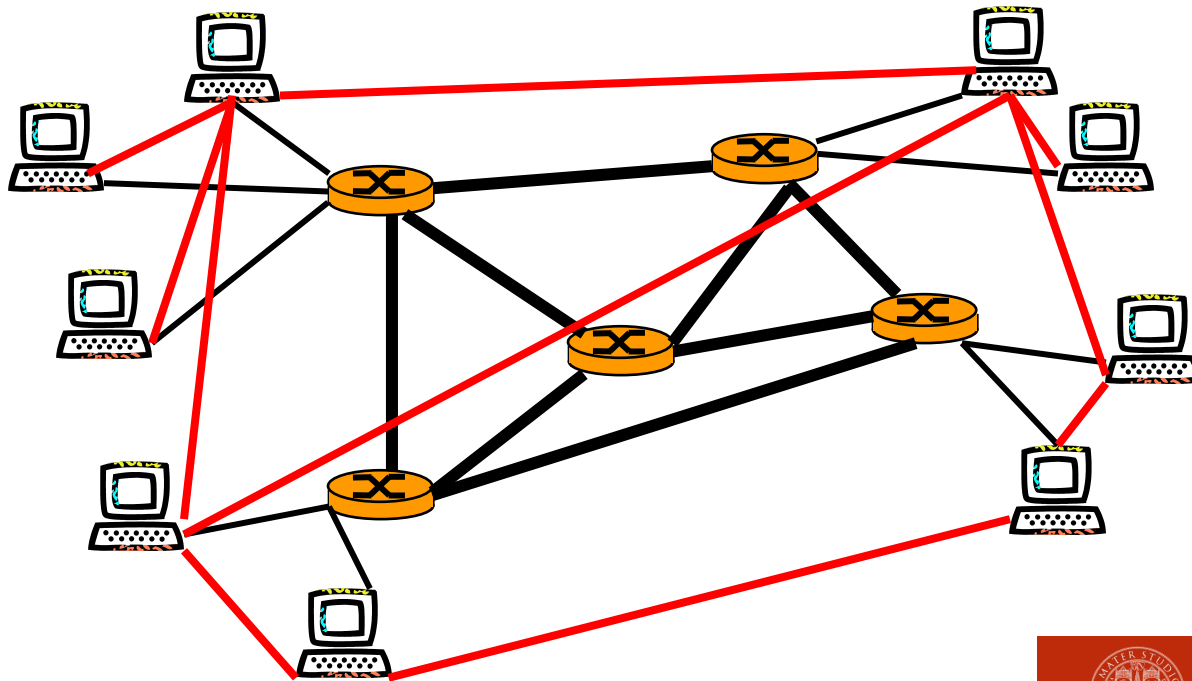
"The miracle is this: The more we share the more we
have." – Leonard Nimoy



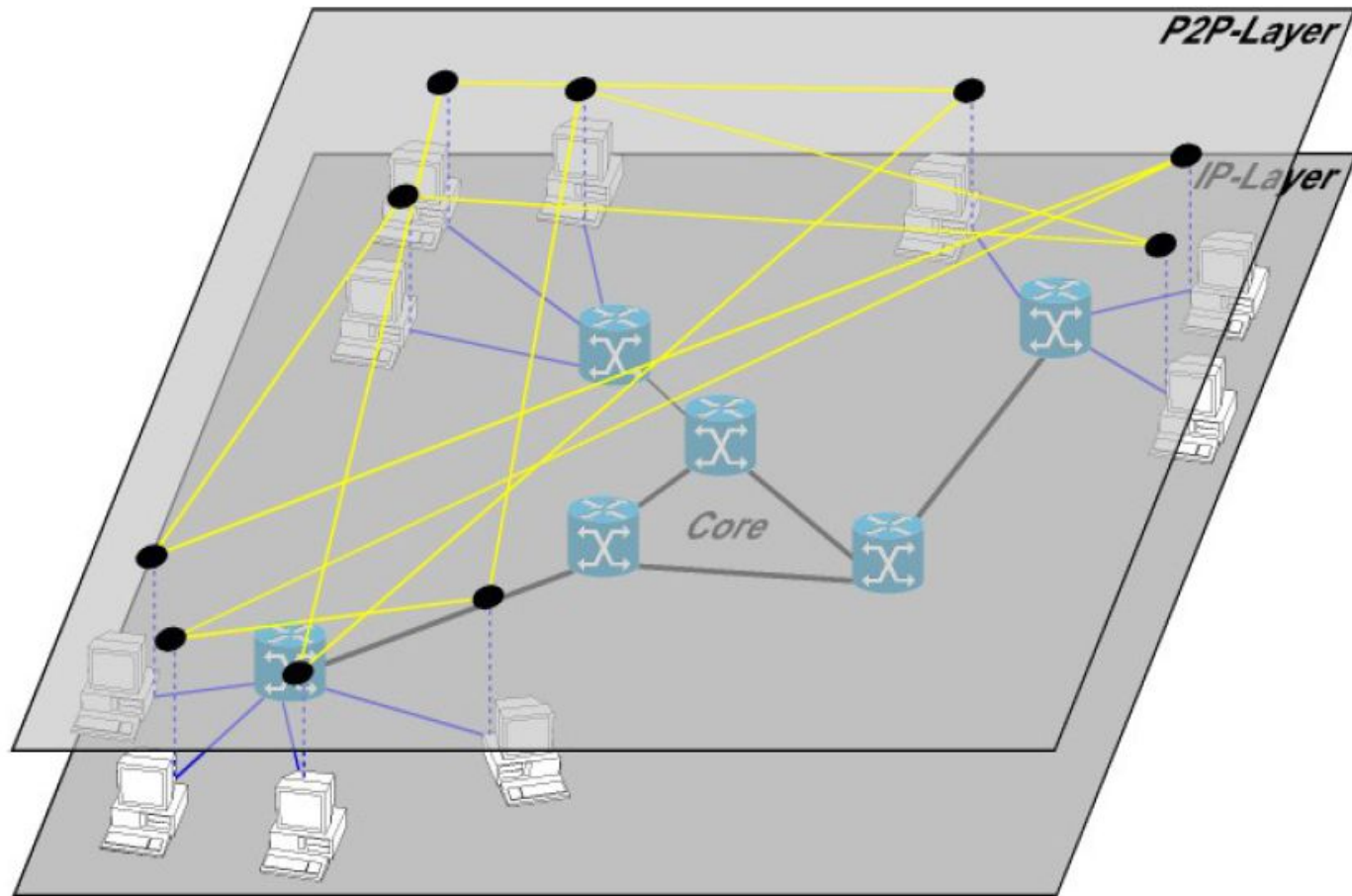
Overlay

□ Overlay Network

- Network at the application layer (layer 7)

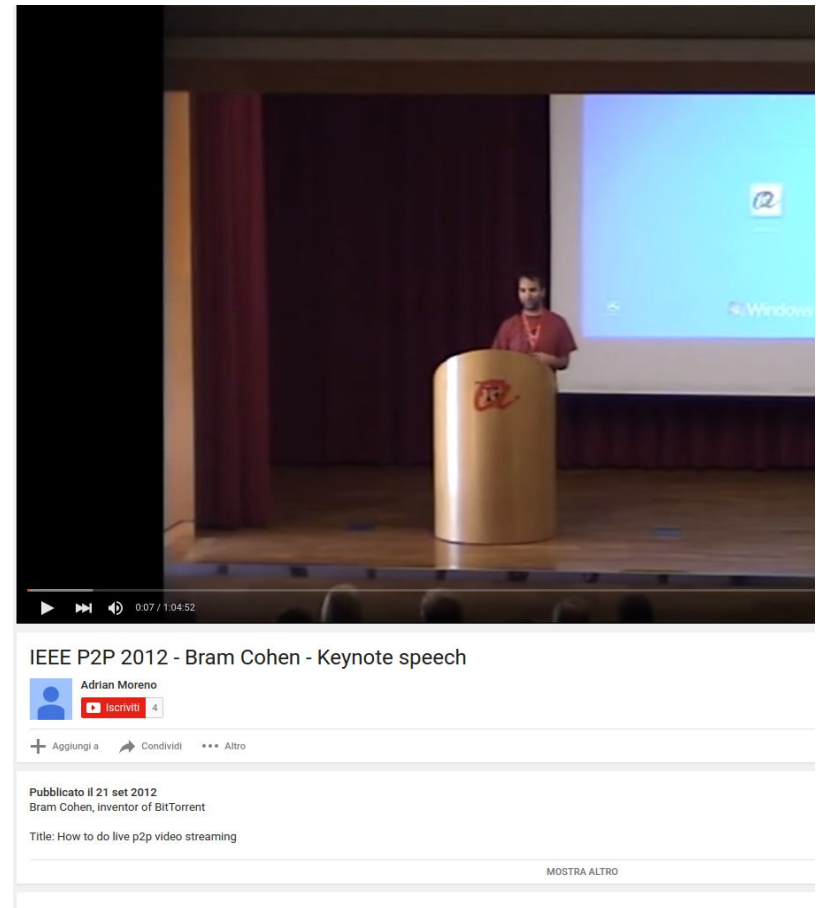


Overlay



Transparency to underlying protocols

- Bram Cohen, inventor of BitTorrent:
- Said to have **no** idea of how TCP works, during the development of BitTorrent
- During a keynote speech @ IEEE P2P 2012
- https://www.youtube.com/watch?v=9_b6n6_yedQ





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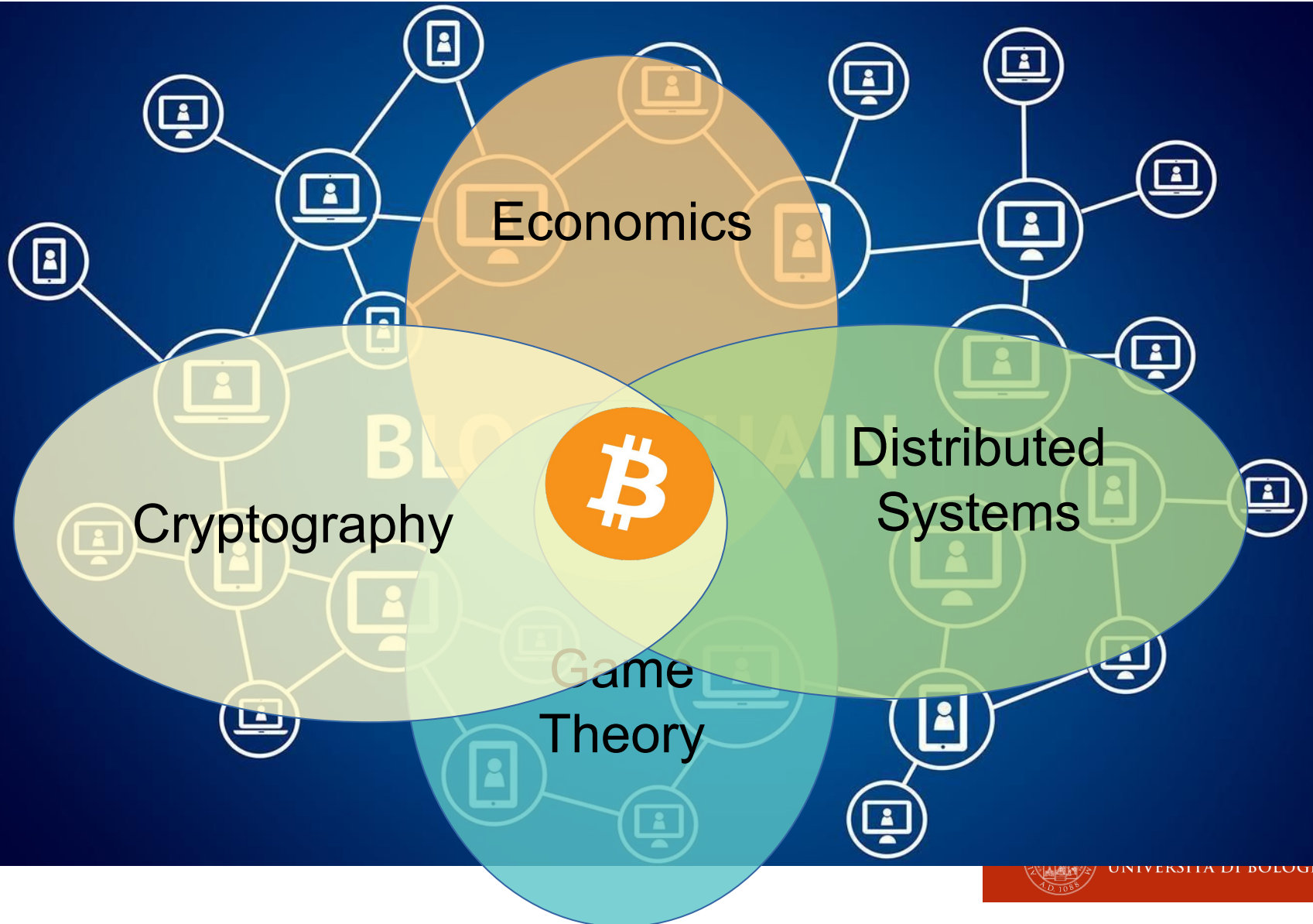
Cryptoeconomics

Cryptoeconomics

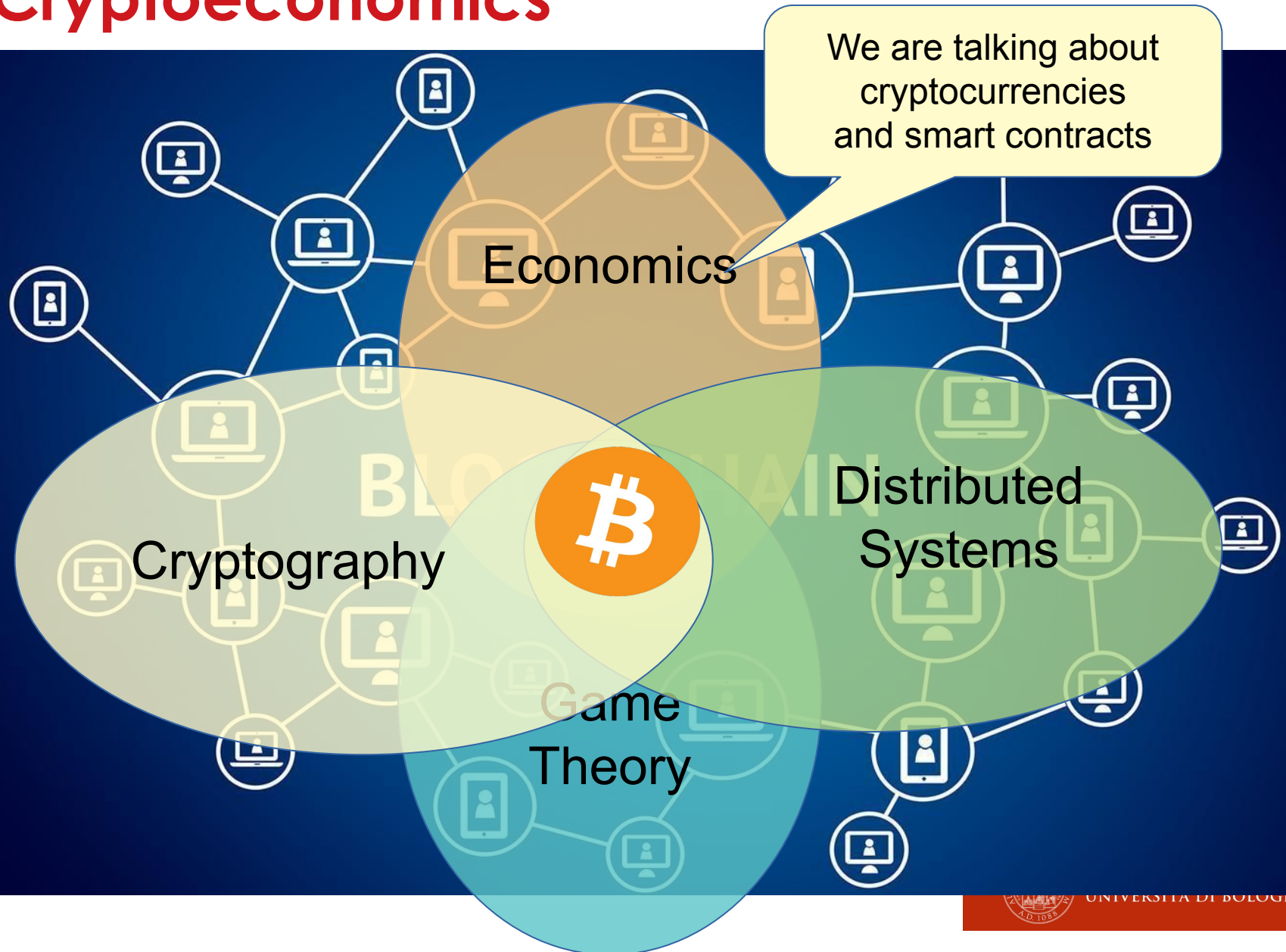
- Cryptoeconomics brings together the fields of economics and computer science to study the applications that can be built by combining cryptography with economic incentives
- It focuses on individual decision-making and strategic interaction between different participants in a digital ecosystem
- The resulting "digital economies" often require the definition of a monetary, fiscal, privacy and innovation policy. Moreover, they need effective governance



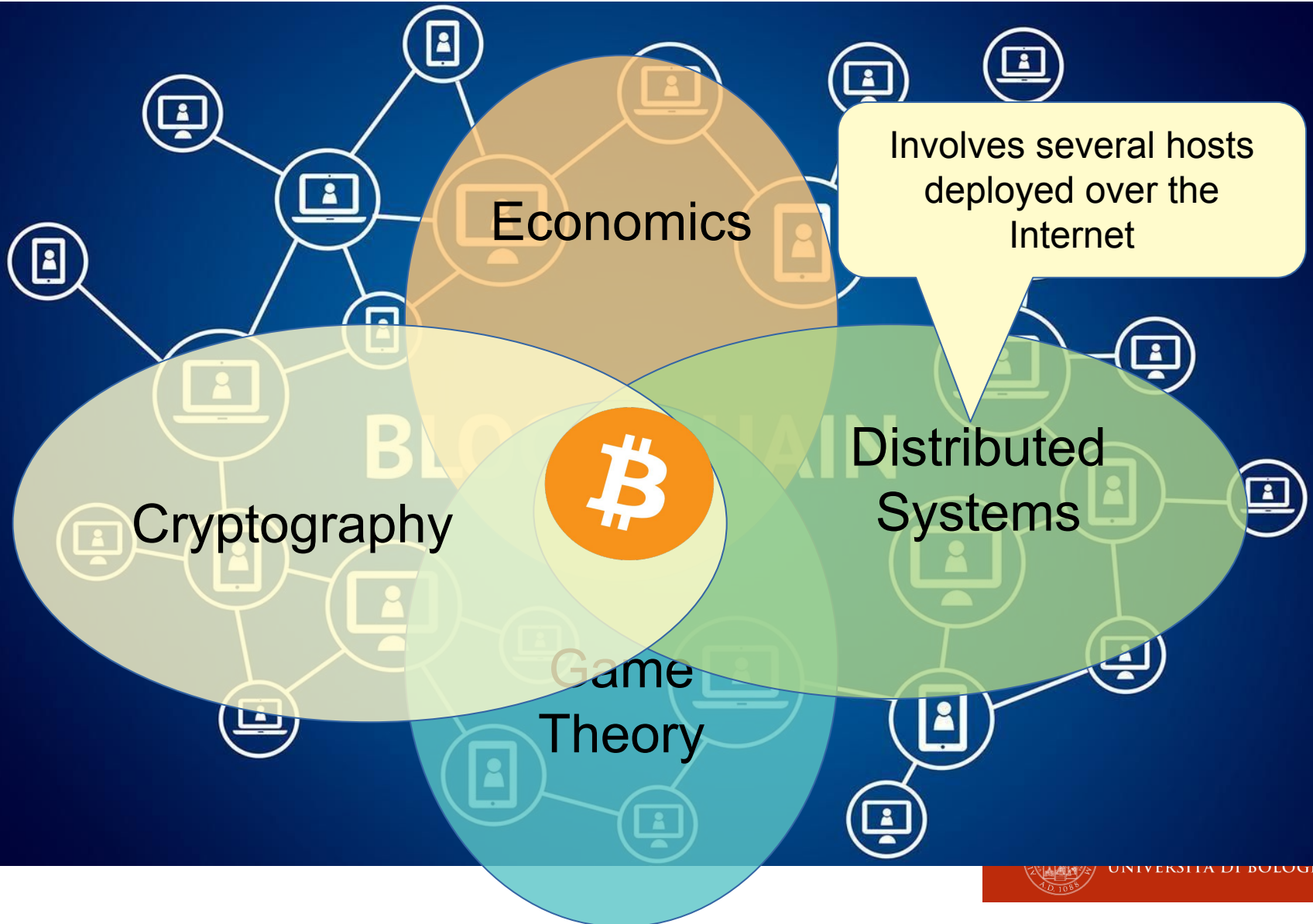
Cryptoeconomics



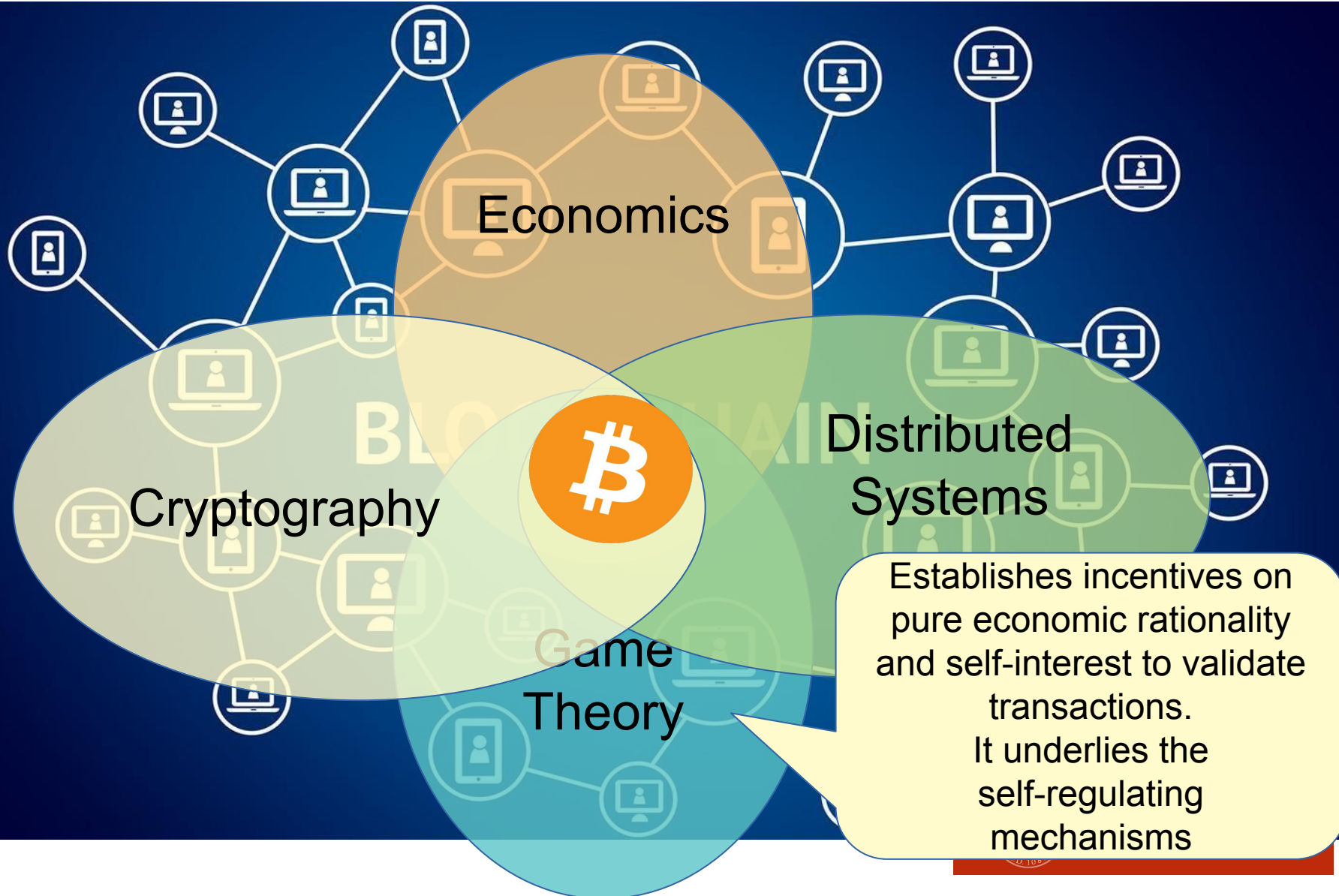
Cryptoeconomics



Cryptoeconomics



Cryptoeconomics



Cryptoeconomics

Essential to provide secure and (pseudo)-anonymous transactions

Economics

Cryptography

Distributed Systems

Game Theory





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Fintech Services

<https://cacm.acm.org/magazines/2018/11/232212-fintech/fulltext>

Service Evolution

Phase 1:
communication

YAHOO!



Service Evolution

Phase 1:
communication

YAHOO!

AOL



Phase 2:
Web 2.0

Google
PayPal

amazon

ebay

Spotify

facebook

Booking.com



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Service Evolution

Phase 1:
communication

YAHOO!

AOL



Phase 2:
Web 2.0

Google
PayPal

amazon

ebay

Spotify

facebook

Booking.com



Phase 3:
share economy

airbnb

Uber



CAR
2GO

GRUPPI
D'ACQUISTO
SOLIDALE:
Strumenti di sobrietà e
buon uso del proprio tempo

DogVacay

kodesk
My office is Your office



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Service Evolution

Phase 1:
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CAR
2GO

Are we sure it is a sharing economy?
It is rather a **matching** economy?

E.g., Uber: you match two users that need to go in near places



kodesk
My office is Your office



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Service Evolution

Phase 1:
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Phase 2:
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Phase 3:
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CAR
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DogVacay

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My office is Your office

Phase 4:
decentralization

ripple

CIRCLE

steemit



blockchain-based
services



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A person is standing on a balcony, seen from behind, with their arms raised in a gesture of freedom or joy. In the background, the Eiffel Tower is visible against a clear sky. The scene is framed by a dark window or balcony railing.

the future
of the

**SHARING
ECONOMY**

WORLD
ECONOMIC
FORUM

<https://www.youtube.com/watch?v=DNBY8yNXGoA>



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Phase 4 Examples: Steemit



trending Nuovo hot promosso

Come funziona Steemit? Mini guida per iniziare! »



alexzicky (60) ▾ in ita • 2 anni fa (edited)

Come funziona Steemit? Bella domanda!

Steemit è un social molto complesso che si basa sulla **blockchain**. Questo significa che tutto quello che postiamo, in nostri **upvote** (like) i commenti e le transazioni (si anche quelle vedremo poi il perché) resteranno indelebilmente scritte nel registro ufficiale di steemit, la blockchain appunto!

Descrizione

Immaginate uno stream come Facebook o Google+ dove si susseguono i post delle persone che seguiamo solo che qui non esiste la pubblicità. In più i guadagni che la piattaforma sviluppa vengono condivisi con tutti i partecipanti, in proporzione alla qualità dei loro post e all'interazione che si ha con gli altri streemers.



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Phase 4 Examples: Ripple



Payments Network ▾ Customers Resources ▾ About Us ▾ Q

Instantly Move Money to All Corners of the World

Discover why 300+ financial institutions choose RippleNet to find new customers in new markets, expand services and deliver the best experience in global payments today.

Join the Network

IndusInd Bank

SABB سائب

SCB
Innovative

Standard
Chartered

Interbank

Ria

Global Payments Should Be Easy



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Phase 4 Examples: Circle



Stablecoins, Global Digital Currency and the Future of the International Monetary System

By Jeremy Allaire, Co-Founder, Chairman and CEO of Circle

January 2020



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Phase 4 Examples: burstIQ



[Home](#) [Developers](#) [Who We Serve](#) [Platform](#) [Company](#) [News](#) [Blog](#) [Conta](#)

A stylized illustration of a human figure, reminiscent of Leonardo da Vinci's Vitruvian Man, rendered in a teal color. The figure is positioned on the left side of a dark blue background. Behind the figure is a complex, circular geometric pattern composed of many small triangles and lines, creating a sense of depth and connectivity. Radiating from the figure's torso are numerous thin, light blue lines that extend across the background, connecting to various points, which visually represents the concept of data connections.

BurstIQ Connects Data Connections Change Health

BurstIQ is a HIPAA and GDPR compliant platform creating health profiles for people, places, and things, and empowers the interactions between them.

[JOIN THE COMMUNITY](#) [CONNECT WITH US](#)



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Phase 4 Examples: mediachain

mediachain 

Building a more connected world for creators and audiences.

Mediachain is a blockchain data solution for connecting applications to media and information about it.

Mediachain uses smart contracts to get musicians the money they deserve.

By entering into a decentralized, transparent contract, artists can agree to higher royalties and actually get paid in full and on time.

Acquired by Spotify in April 2017



Uniting Information

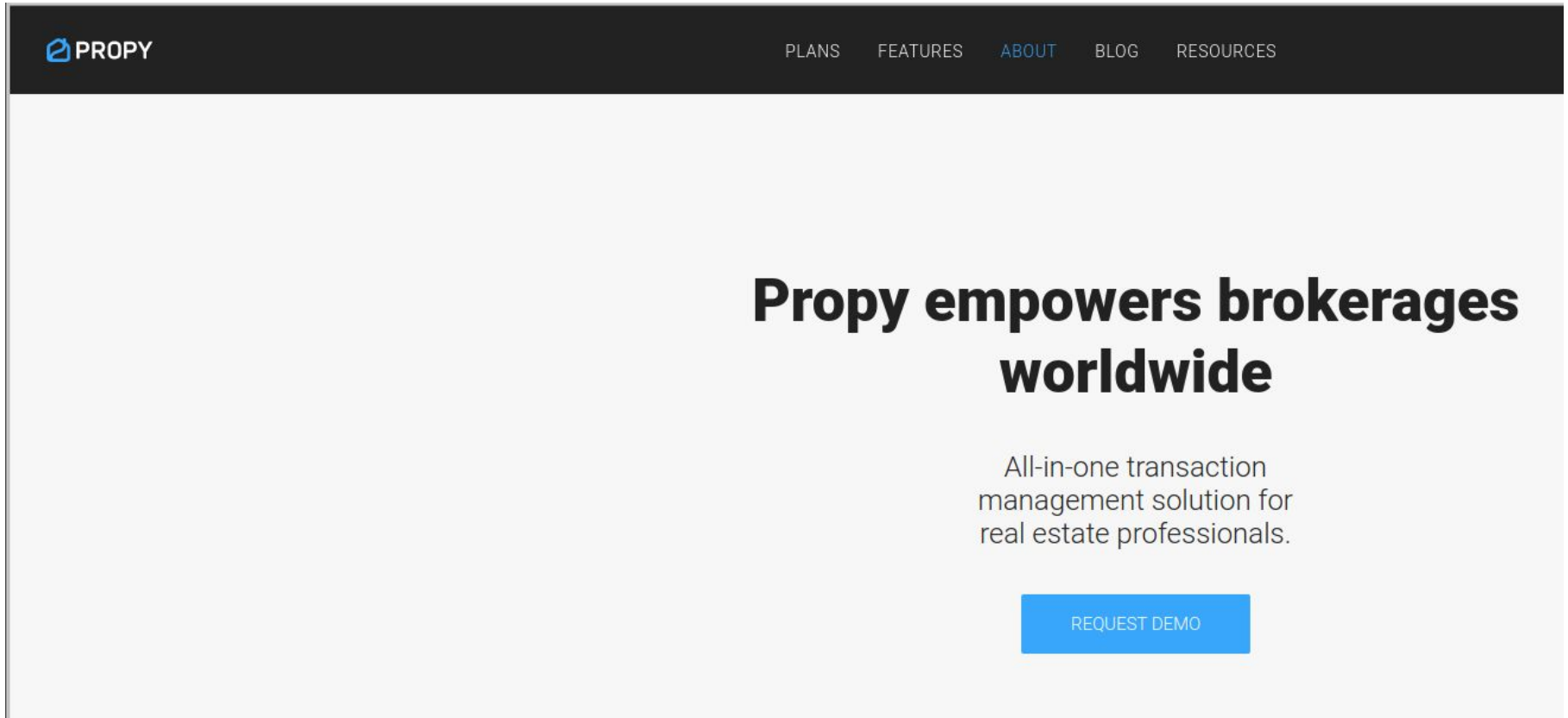
Mediachain is a peer-to-peer, decentralized database for sharing information across applications and organizations.

Using content addressing, data is tamper-proof and decentralized, enabling location-independent collaboration with no central point of control.



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Phase 4 Examples: Propy



Propy is a global real estate marketplace with a decentralized title registry system. The online marketplace uses blockchain to make title issuance instantaneous and even offers properties that can be purchased using cryptocurrency.





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